

# Carlos Moreno

## Curriculum Vitae

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"Yes, if you're looking for infinity, just close your eyes!" – Milan Kundera

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### Personal Information

Name Carlos Andrés Moreno Vélez  
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LinkedIn <https://co.linkedin.com/in/carmoreno1>  
Stack Overflow <https://stackoverflow.com/users/4508767/carmoreno>  
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### Profile

#### Data Engineer, Sr. Python Developer

Carlos has over 9 years of experience in Software Development, with more than 5 of them working with Python technologies as a Data Engineer. Currently, He works as a Scientific Data Engineer at the European Bioinformatics Institute, the company is focused on Bioinformatics and Big Data. He actively participates in developing and maintaining new data pipelines for ChEBI, UniChem and SureChEMBL projects using technologies and languages like Python, SQL, PostgreSQL, Luigi, dbt, PyTest, Polars, Pandas, SQLAlchemy, Bash, Singularity, Docker, Kubernetes, Elasticsearch, Gitlab CI/CD, and SLURM Clusters.

Other technologies he has also worked are Airflow, FastAPI, Django Rest Framework, AWS Redshift, AWS S3, Google Bigquery, Google AppEngine and Google Cloud SQL. He got a Bachelor Degree in System Engineer in 2017 and recently he got a Master Degree in Data Science. He consider himself as a multi-faceted Data Engineer, having experience in Software Development, Data Engineering and a bit of classic ML.

Carlos academic interests include Semantic web, Graph theory and Evolutionary computation.

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### Education

Sep. 2023 - Jan. 2026 **Master Degree - Data Science**, *Open University of Catalunya*.  
Jan. 2021 - Jan. 2022 **EF C1 English Level - Advanced**, *EF English Live*.  
Aug. 2020 - Dic. 2020 **Diploma in Data Science and Machine Learning**, *Universidad Nacional de Colombia*.  
Feb. 2012 - Jul. 2017 **Bachelor of Engineering - Computer Engineer**, *Universidad del Valle*.

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### Experience

Jun. 2022 - Current **Scientific Data Software Engineer**, *EMBL-EBI*, Cambridge, United Kingdom, *Chemical Biology Services*.

- Successfully facilitated the transition of the Release Strategy from custom shell scripts, previously managed through cron to a modern GitLab CI/CD workflow.

- Successfully facilitated the transition of the Release Strategy from custom shell scripts, previously managed through cron, to a modern GitLab CI/CD workflow utilizing Job Schedules.
- Implemented a daily indexation process to map the ChEBI Ontology in an Elasticsearch database with around 200k documents.
- Implemented the new Data pipeline to generate the ChEBI Ontology, using TSV templates extracted from a PostgreSQL database and used by ROBOT (<https://robot.obolibrary.org>) to generate new semantic products.
- Improve data observability for SureChEMBL, automating the release process through Gitlab CI.
- Currently working in the new Unichem Loader, a process which manages more than 400GB of data using gold standard data engineering approaches and tools.
- Participated actively in a data pipeline to migrate more than 14 million records from an ORACLE database to PostgreSQL.
- **Language and tools used:** Python 3.11, Pytest, Pandas, Docker, PostgreSQL, Oracle, Gitlab CI, Elasticsearch, SLURM, Luigi, dbt

Dic. 2020 - May. 2022 **Python Developer Ssr. Advanced**, *Globant*, Cali, Colombia, *Data and AI Studio*.

- Collaborated on a multi-disciplinary team of developers, testers, data architects and devops in order to create and maintain new reports and datapipelines for Education information analysis.
- Built and maintain ETL processes for gigabytes of data about Students, Teachers, US Districts and Schools
- Led 2 initiatives: Data Quality and Database Migrations to improve the development experience in the team and avoid the errors related to SQL dependencies.
- Coordinate meetings with team members from India and LATAM and participate actively in the Code Review process.
- **Language and tools used:** AWS (Redshift, S3, Datapipeline), Python 3.8, Shell, PyTest, Git, Pandas, Flyway, Github Actions, Airflow, Jupyter Notebook, PostgreSQL, Jira.

Feb. 2020 - Nov. 2020 **Python Developer**, *Servinformación*, Cali, Colombia, *Backend Team*.

- "Data Rutas Project": Developed new REST endpoints using a framework legacy version (Django Rest Framework v1.4).
- "Servihuella Project": Integrated data from different sources in collaboration with the Secretary of Health and Mayor of Santiago de Cali. Worked with a total of 5 members on the team among Developers, Stakeholders and Managers.
- "Servihuella Project": Programmed the ETL process of daily data related to COVID-19 using Big data technologies, this ETL processed an average of 2000 rows each day.
- **Language and tools used:** GCP (Big Query, Cloud SQL, Lambda Functions, App Engine, Cloud Scheduler) Python 2.7 and 3+, Django Rest Framework v1.4 (legacy version), CherryPy, Pandas, PyQt, PostgreSQL, MySQL, Git.

Dec. 2018 - Oct. 2019 **Software Engineer**, *El País Newspaper*, Cali, Colombia, *Technology Department*.

- "Digital Subscriptions Project": Developed new REST Endpoints to integrate with Subscriptions providers like Piano (<https://piano.io/>). Additionally, participated in the configuration of the rules in the El País website (<https://www.elpais.com.co>) to implement the Digital Subscriptions, this project works for more than 24.000 subscribers daily.
- "Digital Subscriptions Project": Produced a new web application to use as a back-office, in order to sell Digital Subscriptions, it was used by the selling team (three members at that moment).
- "Digital Subscriptions Project": Communicated with different providers members of Piano to resolve doubts about their product, via email or online conferences. Additionally, communicated with internal Devops members to resolve network errors on the website.
- **Language and tools used:** Javascript, AngularJS, Git, HTML5, CSS3, Bootstrap, PHP, Lumen.

Dec. 2017 - Nov. 2018 **Django / Angular Developer**, *Markaf Colombia*, Tuluá, Colombia, *Development Team*.

- "Markaf App Project": Modeled the internal pay schema of the application: Tables, relations among tables and triggers.
- "Markaf App Project": Developed new REST endpoint according to stakeholders requests.
- "Markaf App Project": Participated in frontend tasks, developed Angular visual components according to UI/UX team needs.
- **Language and tools used:** Python, Django Rest Framework, Angular5, Git, PostgreSQL.

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## Awards

### Laureate Degree Thesis.

Carlos received the Laureate Degree for the bachelor thesis titled "Creation of Dataset about Ecotourism Information of the Center-Region of Valle del Cauca to publish and link with the Data Web". In this job was applied Semantic Web concepts to model data from different sources and formats, linking these resources with others available on the Web. Finally, the created dataset was used in a web application prototype (Semantic REST) in order to show the information to final users. **Language and tools used:** RDF, SparQL, Python, AngularJS, Fuseki.

### Exemption from enrolment fees.

Carlos got a total of five scholar fee exemptions during his bachelor because of academic excellence. The above meant a reduction in the enrolment payment in five different semesters.

### UOC Scholarship for Colombian Students.

UOC scholarship for Colombians students is a program created by the Universitat Oberta de Catalunya which offers economic help to a select group of students because of their academic excellence. Carlos won one out five Master offered scholarships. More info here: <https://www.uoc.edu/es/estudios/becas-ayudas/becas-maestrias-colombia>. **Language and tools used:** Python, SQL, Graph Databases, Neo4j